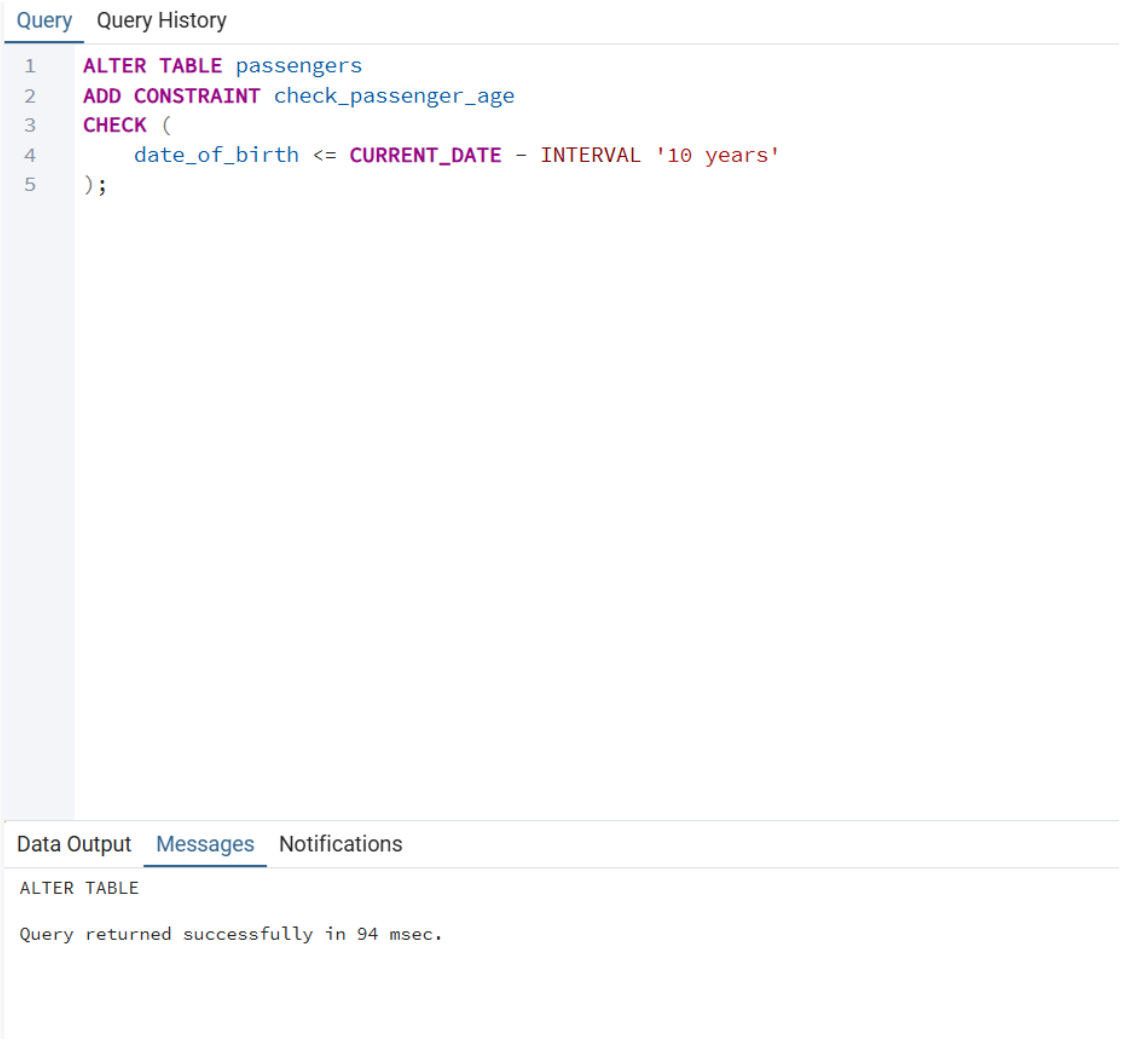
LABORATORY WORK 5

This report documents the SQL tasks completed in the airport_db database using pgAdmin. The screenshots show the executed SQL statements and successful results where available.

Task 1 - Minimum passenger age

Adds a CHECK constraint requiring passengers to be at least 10 years old, based on date_of_birth.

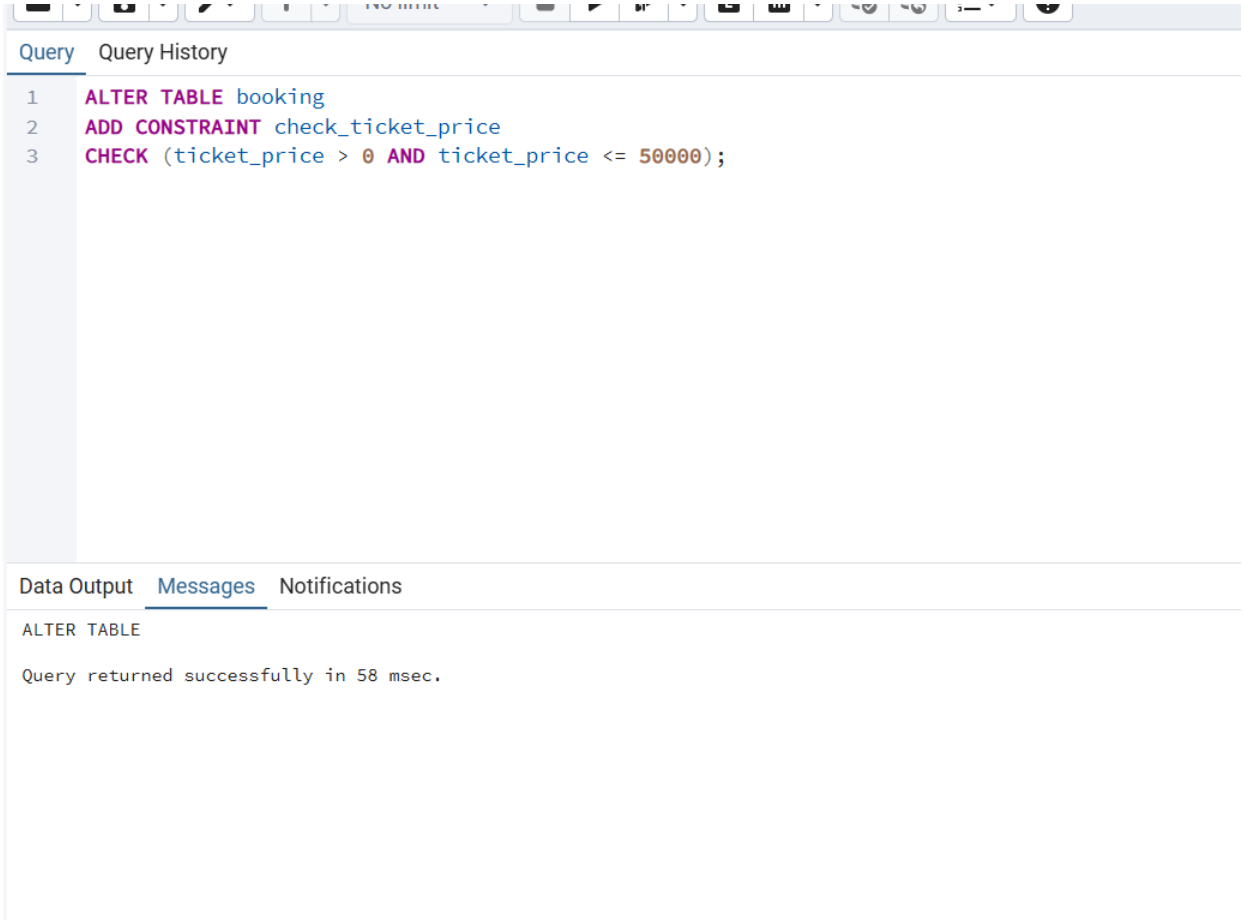
```
ALTER TABLE passengers
ADD CONSTRAINT check_passenger_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
```



Task 2 - Ticket price range

Adds a CHECK constraint requiring ticket_price to be greater than 0 and no more than 50,000.

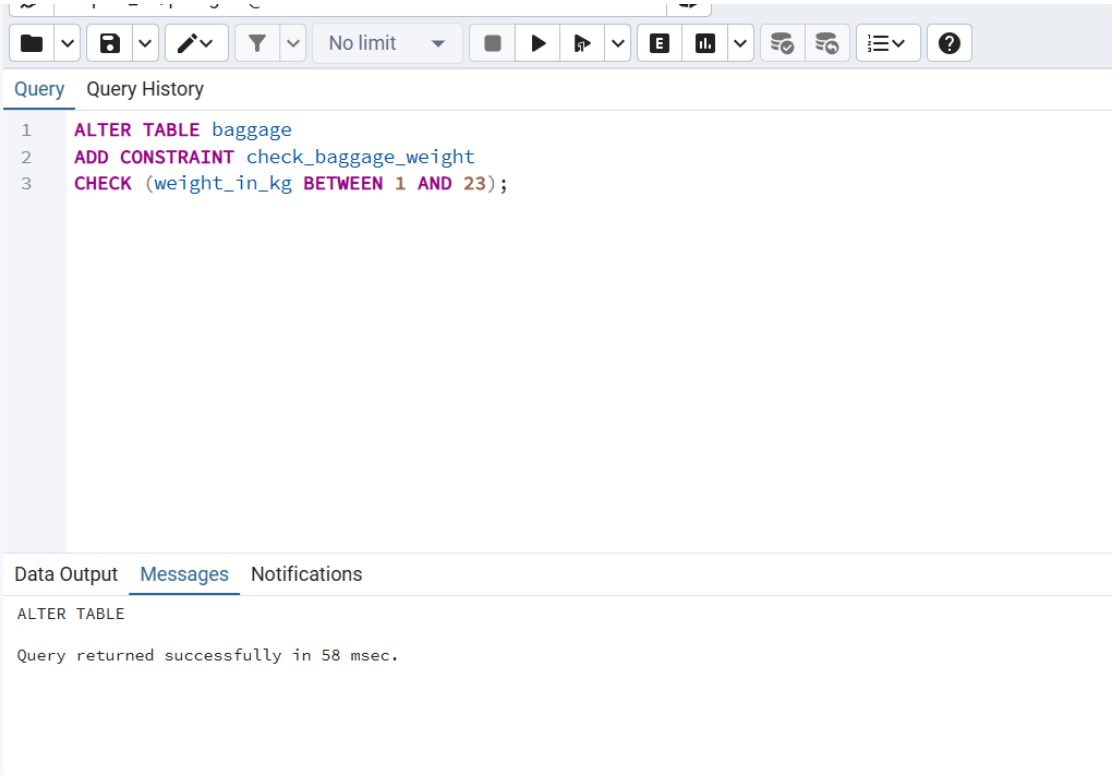
```
ALTER TABLE booking
ADD CONSTRAINT check_ticket_price
CHECK (ticket_price > 0 AND ticket_price <= 50000);
```



Task 3 - Baggage weight

Adds a CHECK constraint requiring baggage weight to be between 1 and 23 kg. An existing 23.40 kg row was changed to 23.00 kg first.

```
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
```



Task 4 - Airport name length

Adds a CHECK constraint requiring airport_name to contain at least 10 characters.

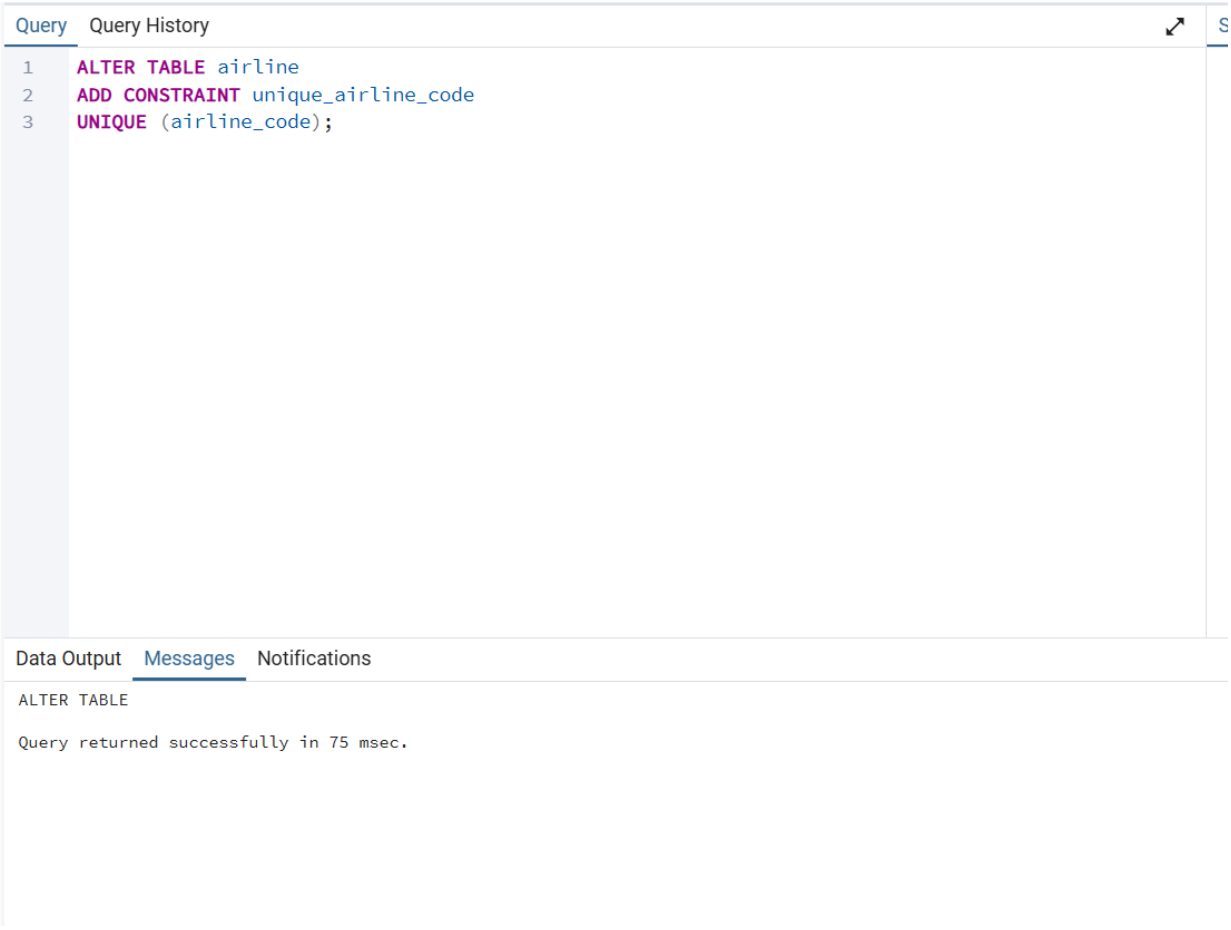
```
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (CHAR_LENGTH(airport_name) >= 10);
```



Task 5 - UNIQUE airline code

Adds UNIQUE (airline_code) to the airline table after checking for duplicate codes.

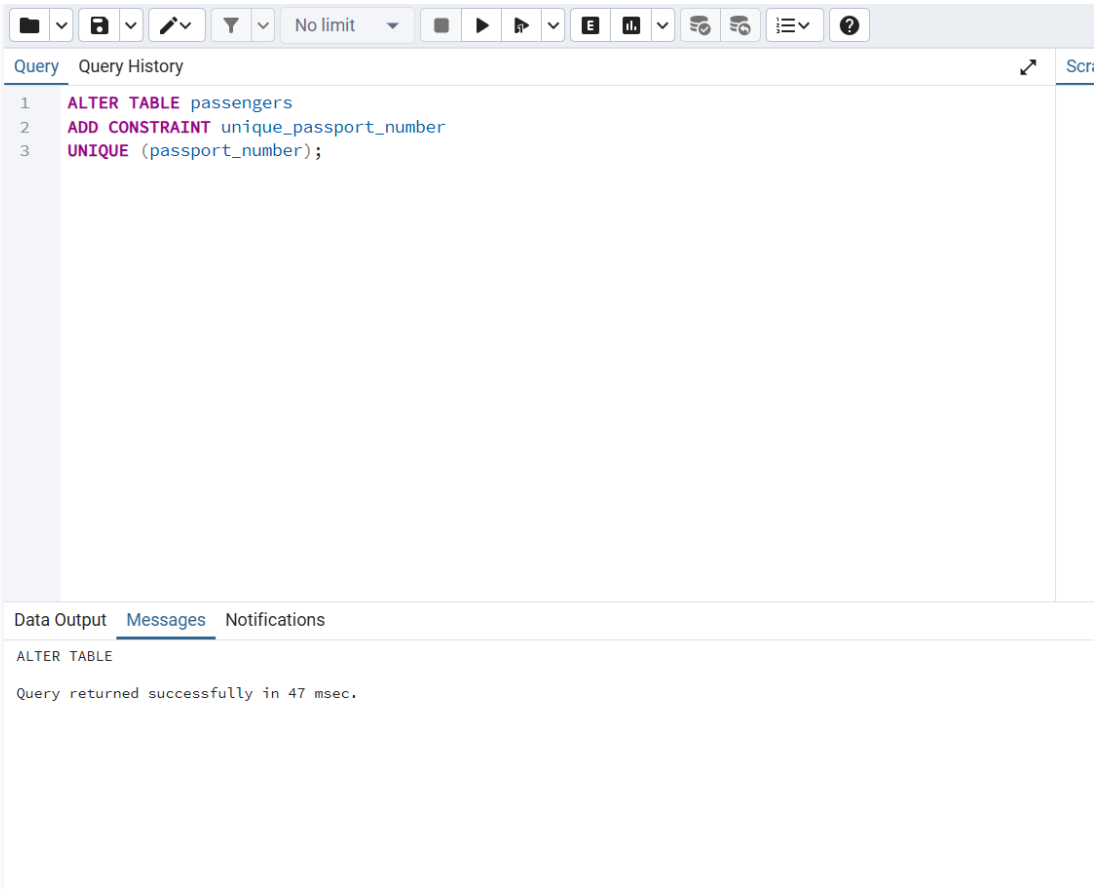
```
ALTER TABLE airline
ADD CONSTRAINT unique_airline_code
UNIQUE (airline_code);
```



Task 5 - UNIQUE passport number

Adds UNIQUE (passport_number) to the passengers table after checking for duplicates.

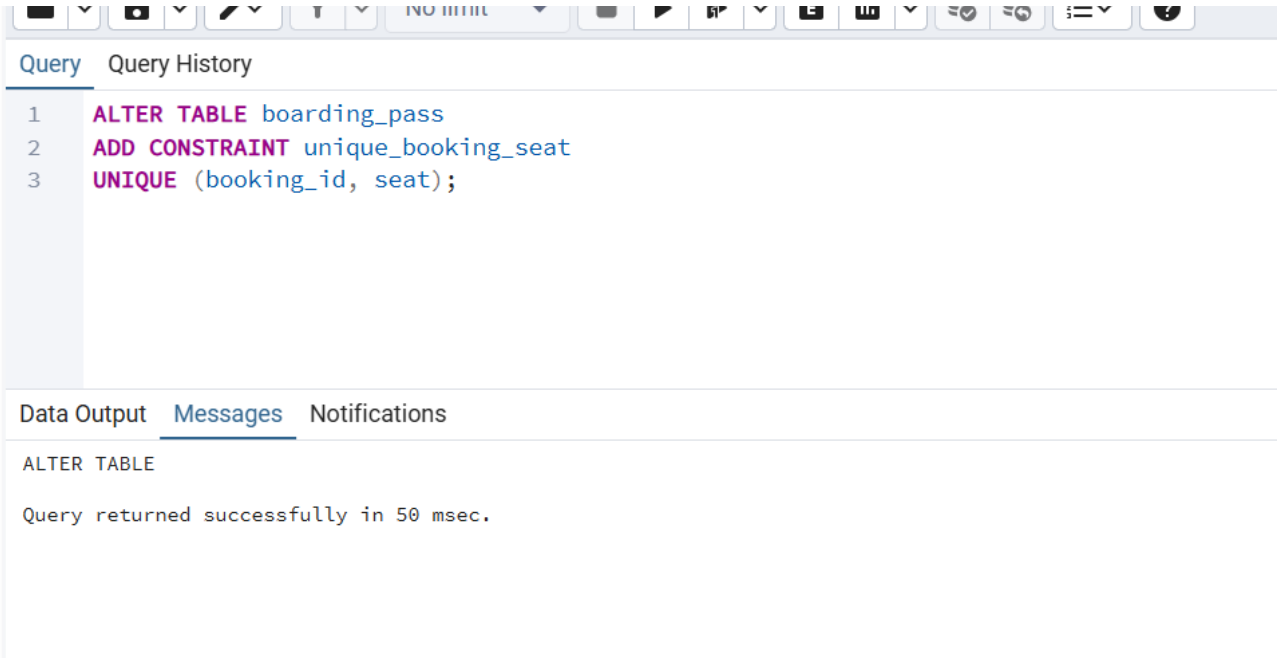
```
ALTER TABLE passengers
ADD CONSTRAINT unique_passport_number
UNIQUE (passport_number);
```



Task 5 - UNIQUE booking and seat

Adds a composite UNIQUE constraint on (booking_id, seat) in boarding_pass.

```
ALTER TABLE boarding_pass
ADD CONSTRAINT unique_booking_seat
UNIQUE (booking_id, seat);
```



Task 6 - Minimum age by gender

Adds a CHECK constraint: Male passengers must be at least 18, and Female passengers at least 19.

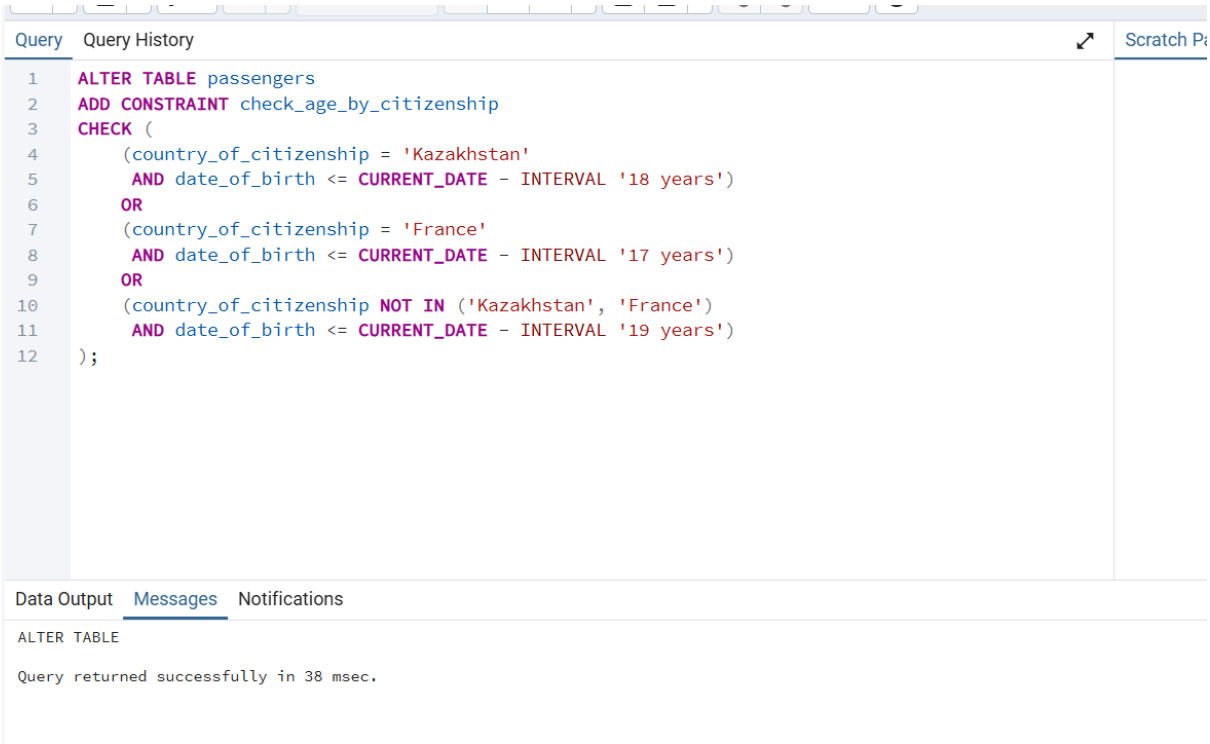
```
ALTER TABLE passengers
ADD CONSTRAINT check_age_by_gender
CHECK (
  (gender = 'Male' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
  OR
  (gender = 'Female' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
```



Task 7 - Minimum age by citizenship

Adds a CHECK constraint: Kazakhstan — at least 18; France — at least 17; other countries — at least 19.

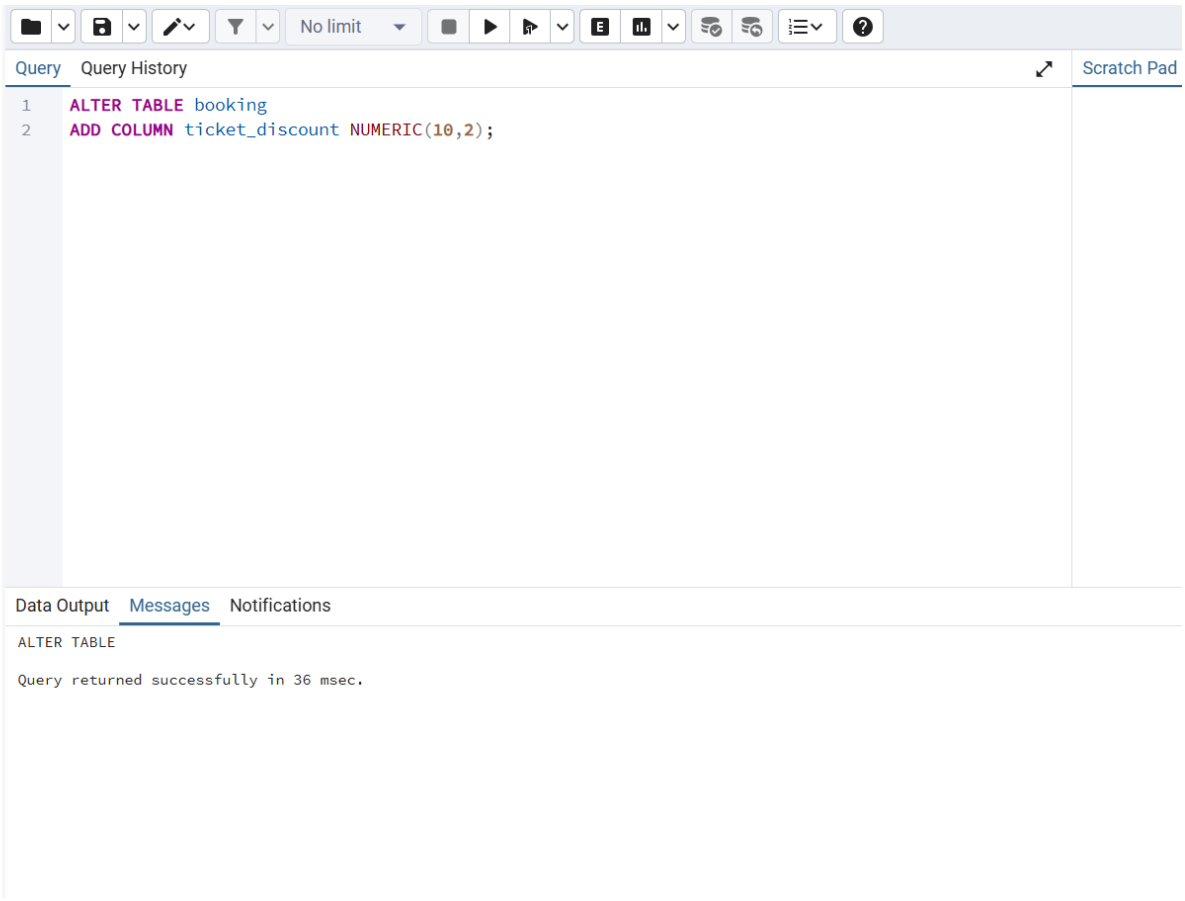
```
ALTER TABLE passengers
ADD CONSTRAINT check_age_by_citizenship
CHECK (
  (country_of_citizenship = 'Kazakhstan' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
  OR
  (country_of_citizenship = 'France' AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
  OR
  (country_of_citizenship NOT IN ('Kazakhstan', 'France') AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
```



Task 8 - Discount column

Adds the ticket_discount column to booking.

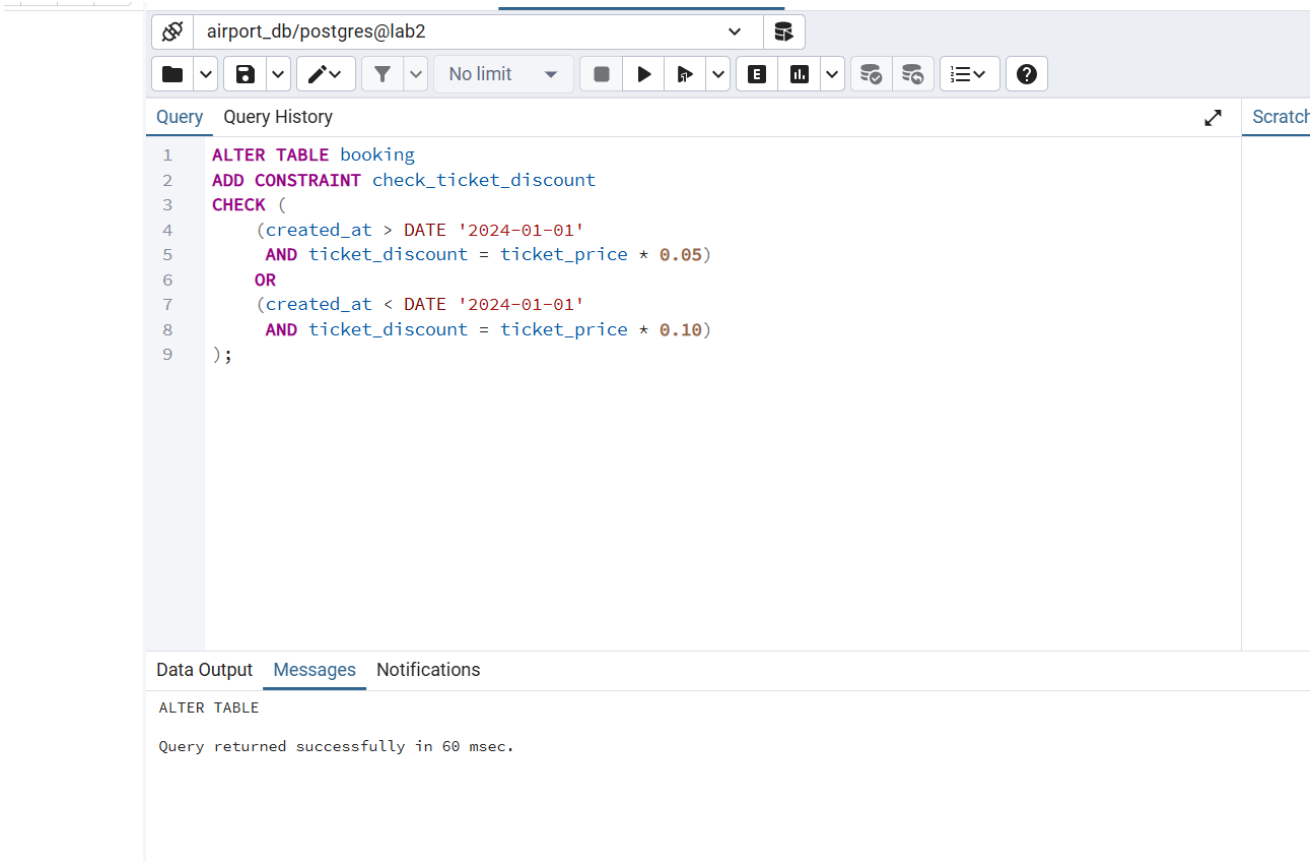
```
ALTER TABLE booking
ADD COLUMN ticket_discount NUMERIC(10,2);
```



Task 8 - Discount rule

Adds a CHECK constraint for 5% discounts on bookings after 2024-01-01 and 10% before that date.

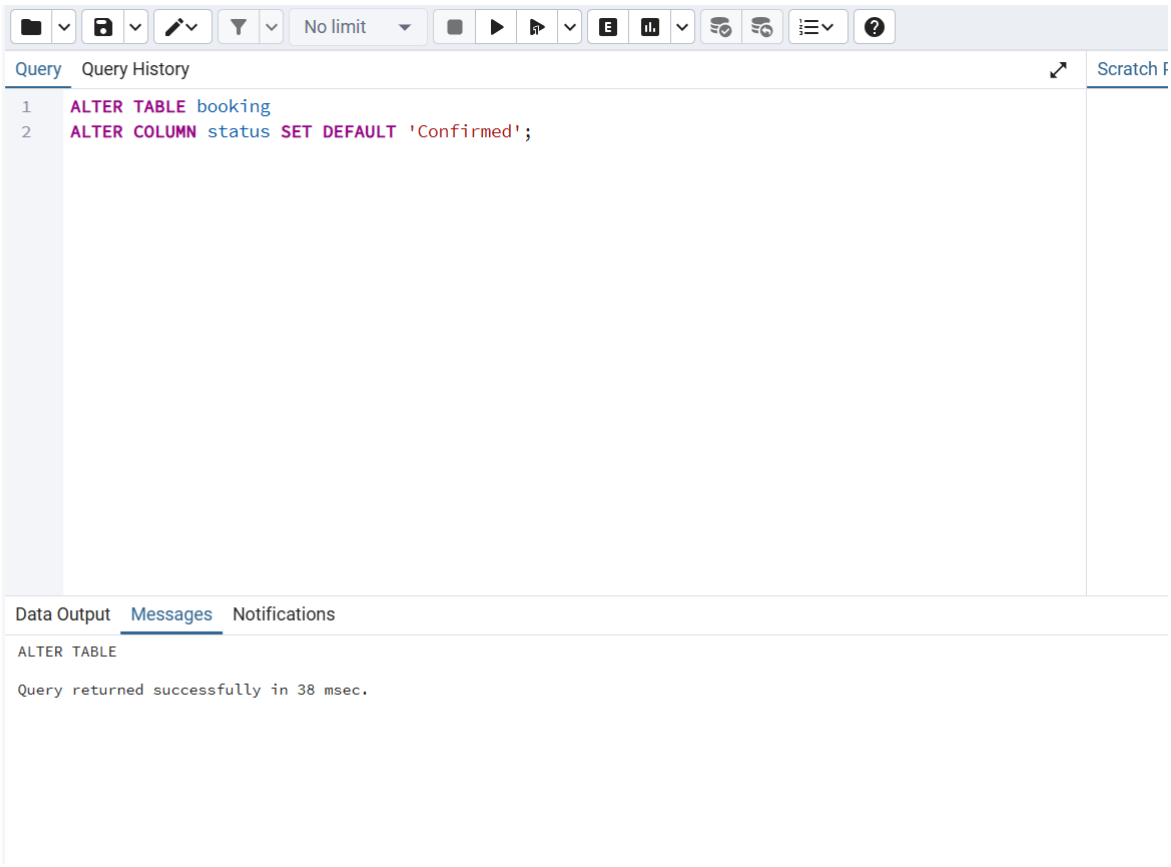
```
ALTER TABLE booking
ADD CONSTRAINT check_ticket_discount
CHECK (
  (created_at > DATE '2024-01-01' AND ticket_discount = ticket_price * 0.05)
  OR
  (created_at < DATE '2024-01-01' AND ticket_discount = ticket_price * 0.10)
);
```



Task 9 - Default booking status

Sets the default value of booking.status to 'Confirmed'.

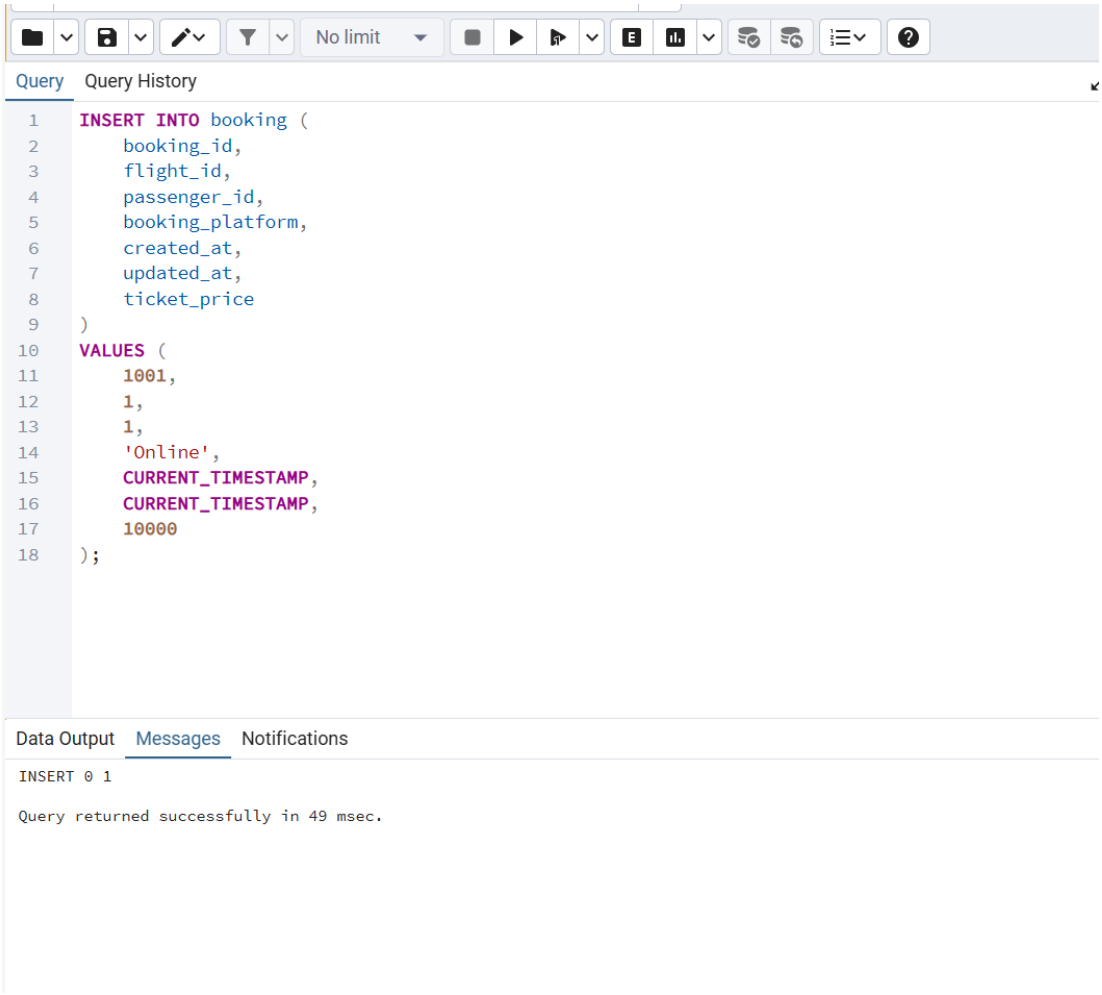
```
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
```



Task 9 - Insert without status

Inserts a booking without specifying status; PostgreSQL reports INSERT 0 1.

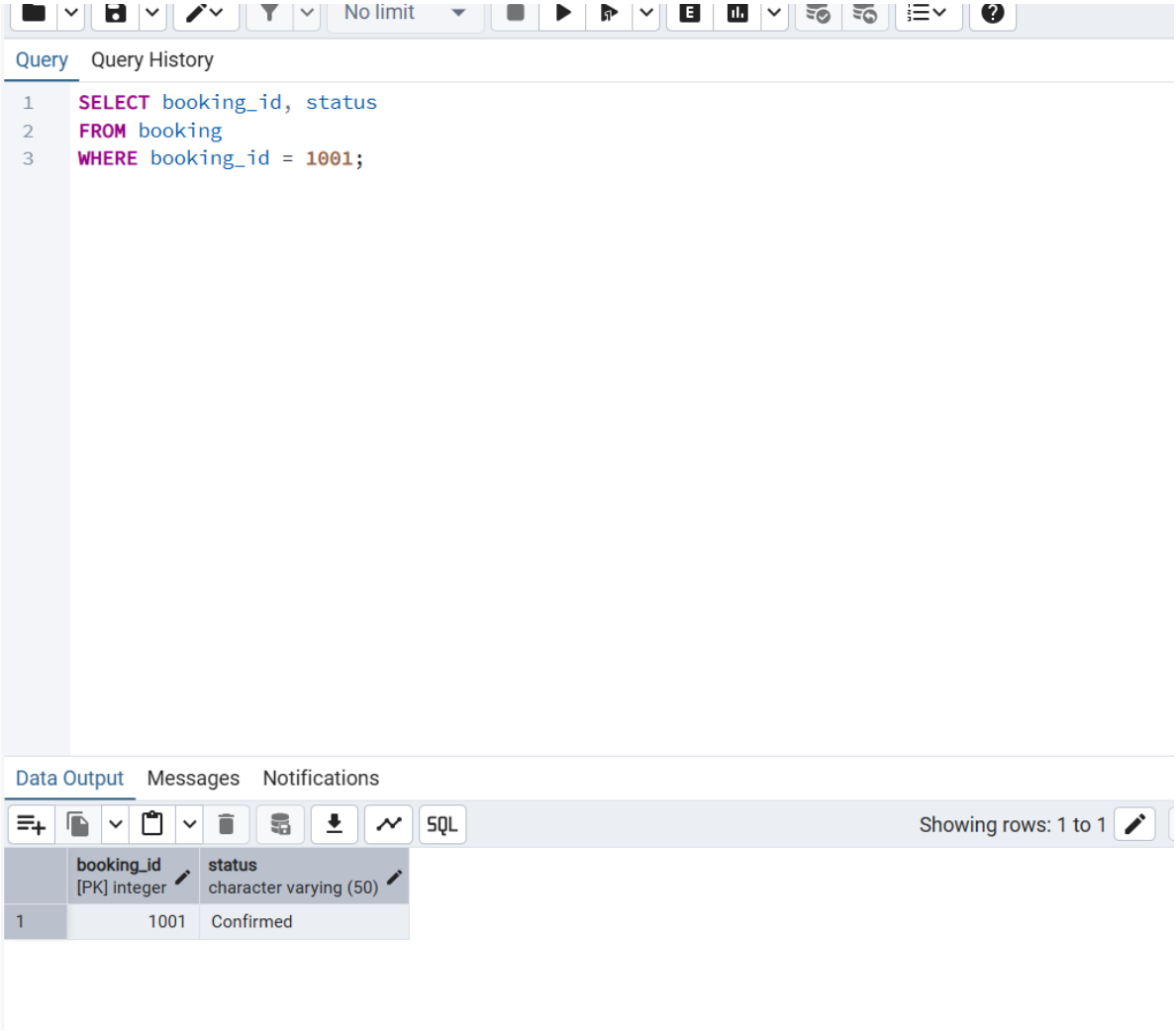
```
INSERT INTO booking (
  booking_id, flight_id, passenger_id, booking_platform,
  created_at, updated_at, ticket_price
)
VALUES (
  1001, 1, 1, 'Online', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 10000
);
```



Task 9 - Verify default status

Confirms booking_id 1001 received status Confirmed automatically.

```
SELECT booking_id, status
FROM booking
WHERE booking_id = 1001;
```



Task 10 - Default created_at

Sets booking.created_at to default to CURRENT_TIMESTAMP.

```
ALTER TABLE booking
ALTER COLUMN created_at SET DEFAULT CURRENT_TIMESTAMP;
```

